

Serum enterolactone and risk of breast cancer: a case-control study in eastern Finland.



Phytoestrogens have been linked to a risk of breast cancer. The main phytoestrogens in the Finnish diet are lignans, and enterolactone is quantitatively the most important circulating lignan. The purpose of this study was to examine the association between serum enterolactone and risk of breast cancer in Finnish women. The subjects were participants of the Kuopio Breast Cancer Study: This analysis concerns 194 breast cancer cases (68 premenopausal and 126 postmenopausal) who entered the study before diagnosis and 208 community-based controls. They completed a validated food frequency questionnaire referring to the previous 12 months and gave serum samples before the examinations. The measurement of serum enterolactone was performed by time-resolved fluoroimmunoassay. The statistical analyses were done by the logistic regression method. The mean serum enterolactone concentration was 20 nmol/l for the cases and 26 nmol/l for the controls ($P = 0.003$). The mean serum enterolactone concentration in the lowest quintile was 3.0 nmol/l and 54.0 nmol/l in the highest. The odds ratio in the highest quintile of enterolactone values adjusted for all of the known risk factors for breast cancer was 0.38 (95% confidence interval, 0.18-0.77; P for trend, 0.03). The inverse association between serum enterolactone and risk of breast cancer was seen both among premenopausal and postmenopausal women. High enterolactone level was associated with higher consumption of rye products and tea and higher intake of dietary fiber and vitamin E compared with those with low serum enterolactone values. Serum enterolactone level was significantly inversely associated with risk of breast cancer.